

When is anti-holomorphic dependence forced?

Exchange non-Hermiticity of one-way transfer kernels and a proposed time-domain test in a recirculating fiber loop

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Abstract

We investigate when the anti-holomorphic (conjugate-variable) dependence of a complex field or kernel is forced by the underlying physics, as opposed to being removable by a choice of gauge, basis, or preparation. As a mathematical result, we show that for two counter-winding mode ladders coupled unidirectionally — via reservoir-engineered, Lindblad-type coupling with a Jordan-block (double-pole) structure per mode pair and a linear frequency ladder — the two-point transfer kernel sums to a closed form involving the Lerch transcendent in the product of conjugated coordinates. Reciprocity is equivalent to the existence of an exchange-Hermitian mirror partner of the kernel; unidirectionality removes this partner. All symbolic claims are certified by exact Wirtinger-derivative computation. We then propose an implementation in a recirculating fiber loop with an intracavity amplifier and an in-loop isolator, together with a time-domain protocol based on a secular revival envelope, a revival comb, and a single-versus-twin-comb reciprocity test. Numerical simulations with literature-typical component values indicate that the secular exponent remains robust under several realistic noise sources, while revival-comb visibility is significantly more demanding in terms of loop-phase stability. No experimental measurements have been performed; experimental verification remains necessary.

1 Introduction

1.1 The question

A complex field or kernel on a two-dimensional carrier generally depends on both the holomorphic coordinate z and its conjugate $\bar{z}$. In many physical settings the anti-holomorphic dependence is an artifact of description: it can be removed by a gauge transformation, by a change of basis, or by a different choice of preparation or probe, after which the object is holomorphic, or real, or a holomorphic function dressed by a modulus factor. This paper is concerned with the opposite situation: when is the anti-holomorphic dependence *forced* — present in every gauge, every basis, and for every admissible preparation? We treat removability operationally: a structure is removable if some transformation within a declared admissible class (unitary or similarity gauge transformations, basis changes among the physical modes, choices of drive or input port) maps the object to one whose mixed Wirtinger structure is trivial. Forcedness is the failure of every such attempt, and we require it to be certified symbolically rather than argued.

1.2 Why one-point fields are insufficient

Two elementary mechanisms empty the one-point case, and motivate the two-point object studied here. First, reality locking: any physically real scalar observable satisfies a conjugation constraint that ties its anti-holomorphic content to its holomorphic content point by point; the two are mirror images

and carry no independent information. Second, preparation erasability: even for a genuinely complex field obtained by driving a system, a one-point snapshot of the driven response depends on the drive, and there exist admissible drives that erase its anti-holomorphic content entirely; structure that can be switched off by an input choice is not forced. Both mechanisms leave a natural escape: an object that is neither a real observable nor a driven snapshot, namely the two-point transfer kernel of an open, non-reciprocal system — the response at one point to a source at another. A kernel is a property of the dynamical generator, not of any drive; and non-reciprocity, implemented at the level of the generator (Lindblad-type one-way coupling), breaks precisely the conjugation-exchange symmetry that reality locking enforces. The body of the paper makes this escape quantitative.

1.3 Results, and what is not claimed

We establish, at the level of symbolically certified formula structure: (i) a closed form for the two-point transfer kernel of two counter-winding mode ladders under per-pair one-way Jordan coupling — a Lerch transcendent in the product of conjugated coordinates; (ii) a dichotomy — reciprocity is equivalent to the presence of an exchange-Hermitian mirror partner of the kernel, and one-way coupling deletes that partner; (iii) a three-fold classification of the resulting kernel (conjugate-sector entanglement, non-pairing, transcendence) with the reproducing-kernel and paired-sum constructions as certified counterexamples on the reciprocal side. We further propose a fiber-loop implementation and a time-domain protocol, and we analyze its robustness against literature-typical imperfections. We claim no experimental result: no measurement has been performed, all component values are catalogue-typical, and the relation of the kernel to superficially similar arithmetic objects (mock modular forms) is discussed and explicitly not asserted. The contribution, if any, is a sharply posed question, a certified structural dichotomy, and a concrete, falsifiable measurement proposal.

Beyond the forcing question itself, the proposed protocol bears on an open experimental debate in exceptional-point (EP) sensing. The claimed sensitivity enhancement of EP-based sensors has been challenged on noise grounds [1, 2], and fundamental-limit analyses have shown that non-reciprocity — rather than EP proximity per se — is the resource that allows a sensor to exceed the bounds constraining any reciprocal device [3]. Both strands of that debate presuppose the ability to certify, in operation and under realistic noise, (i) that a device is genuinely defective (a Jordan-block degeneracy rather than a near-degenerate diagonalizable pair) and (ii) that it is genuinely non-reciprocal. The time-domain observables proposed here address precisely these two certification tasks at low cost: the secular exponent of the revival envelope is a direct witness of Jordan structure which our simulations indicate is robust to loop-phase jitter and between-shot noise (Section 5), and the single-versus-twin revival comb is a direct witness of broken reciprocity. We emphasize that these are certification tools; no sensitivity-enhancement claim is made or implied.

1.4 Outline

Section 2 constructs the kernel and states the dichotomy. Section 3 situates the object among its nearest neighbors. Section 4 describes the proposed platform and protocol; Section 5 the imperfection analysis. Section 6 discusses assumptions, limitations, open questions, and falsifiability. The certification details, the closed-form derivations, and the simulation parameters are provided as supplementary material in the repository accompanying this deposit.

2 Mathematical structure of the one-way kernel

2.1 Setup

We consider two families of modes with opposite winding on a ring-like carrier: an “upstream” family $\phi_m \propto e^{+im\theta}$ and a “downstream” family $\chi_m \propto e^{-im\theta}$, indexed by $m \geq 1$. In complex coordinates $z = re^{i\theta}$

on (or near) the carrier circle, the upstream family is holomorphic-like (z^m up to a radial profile) and the downstream family anti-holomorphic-like ($\bar{z}^m$). Each pair (ϕ_m, χ_m) is coupled unidirectionally: the downstream mode is driven by the upstream one with amplitude g_m , while the reverse matrix element vanishes identically. Such one-way coupling can be produced by reservoir engineering in the sense of cascaded quantum systems [4, 5, 6]: the vanishing of the reverse element is then a property of the Lindblad generator itself, not of a drive or preparation choice, which is the property we rely on throughout. When the two modes of a pair are degenerate and damped at equal rates — which is automatic if they are frequency-degenerate partners coupled to the same bus — the per-pair dynamical matrix is triangular with equal diagonal entries, i.e., a 2×2 Jordan block: the transfer function from upstream to downstream carries a double pole,

$$G_m(\omega) = \frac{g_m}{(i(\omega - \omega_m) - \kappa/2)^2}. \quad (1)$$

We take the pair frequencies on a linear ladder, $\omega_m = m\delta$ (free spectral range δ), with a common linewidth κ .

2.2 Closed form of the two-point transfer kernel

The object of interest is not a one-point field but the two-point transfer kernel: the response at observation point $\mathbf{r}_1$ to a localized source at point $\mathbf{r}_2$. Writing the mode sums at fixed probe frequency and collecting the winding bookkeeping (downstream factor $\bar{z}_1^m$ at the observation point, conjugated upstream factor $\bar{w}_2^m$ at the source point), the kernel takes the form

$$K(\mathbf{r}_1; \mathbf{r}_2) = \sum_{m \geq 1} g_m x^m, \quad x = \bar{z}_1 \bar{w}_2, \quad (2)$$

convergent for $|x| < 1$ (observation and source inside the carrier circle). The weight ladder determines the analytic class:

- (a) flat weights $g_m = g$ give the geometric (algebraic) kernel $gx/(1-x)$;
- (b) weights g/m give the logarithmic kernel $-g \log(1-x)$;
- (c) the physical Jordan ladder $g_m = k/\lambda_m^2$ with $\lambda_m = -\kappa/2 - im\delta$ gives

$$K = -\frac{k}{\delta^2} \left[\Phi(x, 2, c) - \frac{1}{c^2} \right], \quad c = \frac{\kappa}{2i\delta}, \quad (3)$$

where Φ is the Lerch transcendent; the leading structure is a dilogarithm.

Each closed form was verified against its defining series by partial-sum comparison (forty terms, several sample points, geometric tail bound); we found direct high-order numerical differentiation of the Lerch function unstable and avoided it. We emphasize that case (a) shows the one-wayness of the kernel and its transcendence are independent properties: flat weights produce a one-way but algebraic kernel, and it is the physical double-pole ladder that supplies the transcendental class.

2.3 Reciprocity as an exchange symmetry; the mirror-partner statement

Let S denote the exchange of the two points, $(z, \bar{z}) \leftrightarrow (w, \bar{w})$, and let the mirror map M act on a kernel formula by swapping each variable with its conjugate together with $i \rightarrow -i$. We call a kernel *exchange-Hermitian* when $K = M(S(K))$ up to an additive constant; because logarithmic branches can contribute a piecewise constant, the operationally safe test is the constancy of $\exp(K - M(S(K)))$. Exchange-Hermiticity is the formula-level expression of reciprocity: the response from 2 to 1 is the conjugate of the response from 1 to 2.

We observe the following dichotomy, verified symbolically on all cases above and on control kernels. Reciprocal constructions always produce the mirror partner: the diagonal (reproducing-kernel) sum of a single holomorphic sector, $-k \log(1 - z\bar{w})$, is exchange-Hermitian exactly; the Hermitian two-sector sum (a one-way half plus its mirror image) is exchange-Hermitian with the paired structure explicit; and the kernel $\log(z - \bar{w})$, which resembles two-dimensional chiral propagators, is exchange-Hermitian up to a branch constant ($\exp(\Delta) = -1$) and is therefore compatible with reciprocity. Unidirectional coupling, by contrast, deletes the mirror partner: the one-way kernels of Section 2.2 satisfy no exchange-Hermiticity relation, exactly because the reverse transfer element vanishes and nothing restores the conjugate-exchanged term. We summarize this as: reciprocity is equivalent to the presence of the exchange-Hermitian mirror partner; one-wayness removes it, and the unpaired anti-holomorphic half survives alone.

2.4 Classification of the kernel

Three independent formula-level properties position the one-way kernel. First, cross-entanglement: the mixed logarithmic derivatives $\partial_a \partial_b \log K$ across the two points, evaluated on all four conjugate pairings (z, w) , $(z, \bar{w})$, $(\bar{z}, w)$, $(\bar{z}, \bar{w})$, vanish except on $(\bar{z}, \bar{w})$ — the kernel is entangled between the two points, and specifically in the doubly-conjugated sector, as the winding bookkeeping dictates. (A single-pairing test is insufficient; we test all four.) Second, non-pairing: K is not invariant under the mirror map, and no mirror partner accompanies it, in contrast with every reciprocal control. Third, transcendence: for the physical ladder the closed form is a Lerch transcendent, while the flat-weight control is algebraic; transcendence is an independent axis supplied by the double-pole ladder. We do not attach any interpretation to this classification beyond the statements just made.

2.5 Certification methodology

All symbolic statements above — vanishing or non-vanishing of Wirtinger derivatives, the pairing tests, the exchange-Hermiticity dichotomy including its branch-safe form — are certified by exact symbolic computation of Wirtinger derivatives (treating $z, \bar{z}, w, \bar{w}$ as independent variables), in a fixed verification harness that also runs the control battery (reproducing-kernel decoy, Hermitian paired sum, branch-constant case). Numerical evidence (series verification, finite-difference cross-derivatives on transcendental closed forms) supports but never replaces the symbolic certificates. We adopt this separation — a discoverer may propose, only the symbolic judge certifies — throughout the paper.

2.6 Boundary structure of the kernel

The closed form of Section 2.2 converges on the open unit polydisk $|x| < 1$, $x = \bar{z}_1 \bar{w}_2$, and the proposed protocol reads its boundary values. Three structural facts about this boundary are worth recording; all are verified numerically in the repository accompanying this deposit. First, the boundary is intrinsic and dimensionless: it sits at the pure number $|x| = 1$, fixed by the geometry of the mode sum rather than by any parameter choice (the physical rates κ and δ enter the complex offset c , not the location of the boundary), and the formalism contains no dimensional constant from which any other distinguished scale could be built. Second, the boundary is emergent in the mode ladder: every truncation of the sum to N rungs is an entire function of x with no boundary at all, while the radius at which the truncated sum departs from the closed form approaches $|x| = 1$ monotonically as N grows (at the one-percent level, the departure radius is 0.904 at $N = 10$ and 0.991 at $N = 30$, and lies within 10^{-6} of the boundary beyond $N \simeq 100$). The boundary is therefore a property of the infinite ladder that no finite sub-ladder carries — the domain-level analogue of the fact, used throughout this section, that the one-way structure lives in the full ladder rather than in any single Jordan pair. Third, the closed form continues analytically beyond the boundary, where the defining series diverges: the kernel as a function outlives the kernel as a mode sum, at the price of a branch structure on $x \in (1, \infty)$ whose leading discontinuity is inherited

from the dilogarithm ($2\pi i \ln x$; we verified the dilogarithm discontinuity numerically to nine digits, and state the kernel-specific jump at the level of the Lerch-to-dilogarithm reduction rather than as an independently certified result). We record these facts as a structural characterization only: they sharpen, but do not close, the boundary-limit question of Section 6, since a rigorous treatment of the $|x| \rightarrow 1$ limit in the presence of the complex offset remains open.

Two further structural facts about this boundary were established numerically after the characterization above, and are recorded here with the same status discipline (machine-verified simulation of the kernel; no physical claim). First, the crossing of the boundary is *oriented*: writing the leading discontinuity of the closed form across the cut $x \in (1, \infty)$ as $\text{disc } \Phi(x, 2, a) = 2\pi i x^{-a} \ln x$, the complex offset $a = 1 + \kappa/(2i\delta)$ produced by one-way coupling makes the *phase* of the jump rotate along the cut (by $\arg x^{-(a-1)}$, about 9.2° from $x = 1$ to $x = 5$ at $\kappa/\delta = 0.2$), whereas the reciprocal (Bergman-type, dilogarithm) control, whose offset is real, has a frozen jump phase at every point of the cut. The ratio law $\text{disc}_{\text{one-way}}/\text{disc}_{\text{reciprocal}} = x^{-(a-1)}$ was verified digit-for-digit against the dilogarithm control (itself machine-exact to 10^{-9}); the monodromy stacking across repeated crossings is arithmetic (equal sheet steps, ratios equal to unity to 10^{-21}). We are not aware of a published use of the cut-discontinuity phase law as a reciprocity witness; the closest works we found diagnose directionality through spectral winding numbers or end-to-end Green’s-function growth [23], through the phase of the drive [19], or through a frequency-dependent 2π phase gradient at a branch-cut crossing in non-Hermitian metasurfaces [22]; none states a position-dependent jump phase along the cut, and we flag the usual caveat that niche terminology may hide precedents. Second, when several kernels with different ladder spacings are summed, their tears stack on the same ray and the jumps add linearly, but the phase windings differ, and the resulting interference relief along the cut is governed by a *detuning* law: at fixed spread of ladder spacings, increasing the number of summed kernels averages the phases and flattens the relief, whereas at fixed number, widening the spread of spacings builds it (contrast growing by an order of magnitude and interior interference extrema appearing), while the reciprocal control remains flat at machine precision throughout. Together with the truncation behaviour above, this separates two emergence mechanisms living on the same object: the boundary itself emerges from the *number* of rungs, while structure *along* its tear emerges from their *diversity*. The emergence-with- N pattern of an analyticity boundary has well-known precedents in statistical mechanics — the accumulation of partition-function zeros in the thermodynamic limit [20] and the natural boundary of the 2D Ising susceptibility built from singularities that become dense as the order grows [21] — and we present the truncation ladder of Section 2.6 as an instance of that phenomenon in a two-point transfer kernel with complex offset, not as a new class.

3 Relation to prior structures

None of the ingredients used above is new in isolation; what we believe has no direct precedent is the combination of the forcing question, the mirror-deletion statement, the derived Lerch kernel, and the symbolic certification discipline. This section states, for each neighboring body of work, what is similar and what is structurally different. We make no priority claim beyond “no direct precedent found in our search”; we would welcome pointers to closer antecedents.

3.1 Bergman and Szegő reproducing kernels

Similar: reproducing kernels of holomorphic function spaces are two-point, cross-conjugate objects — functions of $z\bar{w}$ — and include logarithmic forms such as $-\log(1 - z\bar{w})$; at the formula level they occupy the same cross-conjugate class as our kernel and served as our principal controls. *Different*: reproducing kernels satisfy $K(\mathbf{r}_1, \mathbf{r}_2) = \overline{K(\mathbf{r}_2, \mathbf{r}_1)}$ by construction — they are exactly exchange-Hermitian, i.e., they belong to the reciprocal class. In our terms, they always come with their mirror partner. Our one-way kernel differs precisely in this respect: no exchange-Hermiticity relation holds, because the reverse

transfer element vanishes identically. We regard the Bergman kernel as the sharpest decoy for the present construction: a transcendental, cross-conjugate, non-mirror-symmetric formula that is nevertheless fully compatible with reciprocity. The exchange-Hermiticity test was introduced into our battery specifically because a formula-level gate without it accepts this decoy.

3.2 Cascaded quantum systems and reservoir-engineered non-reciprocity

Similar: the coupling mechanism we use is entirely standard. Unidirectional (cascaded) coupling of quantum systems goes back to Gardiner and to Carmichael [4, 5], and the reservoir-engineering recipe that makes one matrix element of a coupled pair vanish while keeping the other finite is that of Metelmann and Clerk [6], with several experimental realizations in superconducting circuits and optomechanics. *Different:* those works engineer and characterize the non-reciprocal coupling itself (isolation, directional amplification, scattering matrices). The added content here is downstream of the mechanism: the classification of the two-point transfer kernel that a whole ladder of such couplings produces (its conjugate-sector entanglement, its non-pairing, its transcendence class), and the statement that reciprocity at kernel level is equivalent to the presence of the exchange-Hermitian mirror partner. To our knowledge the kernel-level dichotomy is not stated in the cascaded-systems literature.

3.3 Mock modular forms and Appell–Lerch sums

Although the closed-form expression of the kernel involves the Lerch transcendent, reminiscent of certain mock-modular forms and Appell–Lerch sums, the object studied here is fundamentally different in both origin and structure. Our kernel arises from a unidirectional Lindblad coupling between counter-winding mode ladders, with a closed form convergent on the unit polydisk; it carries no modular transformation properties, and exhibits a strictly anti-holomorphic dependence on the product of conjugated coordinates. It should therefore not be regarded as a mock-modular form, but rather as an exchange-non-Hermitian one-way transfer kernel.

This is the most delicate comparison, and we treat it explicitly because the same special function appears on both sides. *Similar:* in the theory of mock theta functions as completed by Zwegers [7], and in its subsequent development [8, 9], a holomorphic q -series acquires good (modular) transformation behavior only after the addition of an explicitly non-holomorphic completion term; the building blocks are Appell–Lerch sums, which belong to the same Lerch family as the transcendent in our closed form. There is, moreover, a structural motif shared with our construction: an object that is “one half” of a symmetric pair, whose missing partner would restore a symmetry — modularity there, exchange-Hermiticity here — and whose interest lies precisely in the uncompleted half. *Different, and decisively so at the technical level:* (i) our kernel is a function of a single product variable $x = \bar{z}\bar{w}$ on the unit polydisk, with no modular variable τ , no lattice, no weight, and no transformation law — nothing in our construction transforms under $SL(2, \mathbb{Z})$ or any subgroup; (ii) the anti-holomorphic dependence of a mock form’s completion is arithmetic in origin (an Eichler-type period integral in $\bar{\tau}$), whereas ours is dynamical (winding bookkeeping of counter-propagating ladders under Lindblad-type one-way coupling); (iii) the parameter c in our Lerch transcendent is a dynamical complex offset, $\kappa/(2i\delta)$, not a lattice point; (iv) the symmetry whose breaking defines our object is the exchange of two spatial points, not a modular transformation. We therefore do not claim any correspondence between the two settings. Making one precise would require, at minimum, exhibiting a map under which the exchange operation realizes a modular transformation and the one-way kernel arises as the shadow or completion term of a specific mock object; we do not possess such a map, and we flag its existence or non-existence as an open question (Section 6). The resemblance, as it stands, is a shared special-function family and a shared “missing-partner” motif — not an identity.

3.4 Non-reciprocal lattice models and the Hatano–Nelson asymmetry

Similar: unidirectional dynamics of Hatano–Nelson type — generated by an imaginary vector potential — is known to leave universal, closed-form fingerprints in its counting and echo statistics: in the interacting Hatano–Nelson model, the full counting statistics takes a universal Gaussian form whose mean scales with the imaginary vector potential [10], and the work statistics exhibits a universal high-energy tail with exponent -3 , originating from a temporal power-law decay of a generalized Loschmidt echo [11]. We read this as independent evidence that unidirectional dynamics writes universal structures into its generating functions. *Different:* the observables there are counting-statistics and Loschmidt-echo objects in a one-dimensional lattice, with no two-point spatial kernel, no conjugate-sector classification, and no exchange-Hermiticity statement; conversely, the Hatano–Nelson asymmetry itself is gauge-removable by an imaginary similarity transformation, whereas the one-way coupling used here is Lindbladian and not removable by any such similarity — a distinction that matters for the forcing question and that we test explicitly.

3.5 Exceptional points and double poles

Similar: the per-pair ingredient of our construction — a defective (Jordan) 2×2 block whose transfer function carries a double pole — is standard exceptional-point physics [12], and chiral exceptional points in whispering-gallery resonators are an active experimental subject. *Different:* a single exceptional point yields a single azimuthal Fourier tone of the two-point response and no transcendental structure; in our construction it is the coherent ladder of Jordan pairs, with its double-pole weights summed over the winding index, that produces the Lerch transcendent. We state this as a control result: the EP is necessary per pair but nowhere near sufficient; the ladder is the point.

Time-domain signatures of defective degeneracies also have an experimental precedent: Dietz *et al.* observed the algebraic-times-exponential decay (a t^2 intensity dependence) predicted at an exceptional point of a dissipative microwave billiard [13]. Relative to that precedent, the present proposal differs in the platform (a recirculating fiber loop read on an oscilloscope rather than a microwave cavity), in the observable set (the secular envelope is combined with a phase-swept revival comb and a single-versus-twin-comb reciprocity test, which the billiard setting does not probe), and in the object being certified (the kernel-level deletion of the exchange-Hermitian mirror partner by one-way coupling, certified symbolically upstream). The secular envelope itself is therefore not claimed as new; its use as a cheap, noise-robust certification channel for one-way kernels is.

3.6 Cousins of the forcing question

The question “when is anti-holomorphic dependence forced rather than chosen” has recognizable cousins in at least three settings: the holomorphic anomaly of topological string amplitudes [14], where consistency forces a controlled anti-holomorphic dependence on moduli; tt^* geometry [15], where holomorphic data are completed by a metric mixing the two sectors; and poly-analytic function theory [16, 17, 18], which studies function spaces genuinely dependent on $\bar{z}$, notably in higher Landau levels. *Similar:* in each case a structure that “would like” to be holomorphic is obliged, by consistency or by the physics, to carry anti-holomorphic content. *Different:* the objects (moduli-space amplitudes, ground-state metrics, single-particle function spaces), the mechanisms (anomaly equations, supersymmetric geometry, Landau-level structure), and the notion of removability all differ from ours; none of these settings poses the removability question for a two-point transfer kernel under non-reciprocal open-system dynamics, which is the specific question addressed here.

4 Proposed experimental platform and protocol

4.1 Chiral feedback bus on a recirculating fiber loop

We propose the following implementation. A recirculating fiber loop (length L , free spectral range $\delta = c/(n_g L)$) supports frequency-degenerate pairs of counter-propagating modes, indexed by the longitudinal number m ; on the loop, counter-propagation realizes the counter-winding pairing of Section 2 (the two members of a pair carry azimuthal factors $e^{+im\theta}$ and $e^{-im\theta}$). The loop is coupled to an add-drop bus; the clockwise output of the bus is routed through an optical isolator and re-injected onto the counter-clockwise input. This “chiral feedback bus” has three properties that match the mathematical construction of Section 2 without fine tuning. First, the two members of each pair are degenerate and couple to the same bus with the same rate, so the equal-diagonal condition of the per-pair Jordan block holds automatically. Second, the isolator suppresses the reverse path, so the vanishing of the reverse matrix element is a structural property of the setup rather than a tuned condition. Third, a single feedback loop serves the entire mode ladder simultaneously, with per-rung coupling phases $g_m = g_0 e^{im\phi}$ set by the loop delay ($\phi = \delta$ times the loop delay), which amounts to a rigid rotation of the ladder. Numerically, the ladder sum over this structure reproduces the closed form of Section 2.2 — evaluated on the unit circle of its convergence domain — to a relative accuracy of 1.4×10^{-7} in our implementation, which we regard as an identity check between the abstract kernel and the device model rather than as an approximation.

4.2 Mapped parameters

For concreteness we map the model onto catalogue-typical components (all values to be re-verified against the actual components of any experimental attempt): $L = 20$ m of standard single-mode fiber ($\delta/2\pi = 10.2$ MHz, round-trip time 98 ns); an in-loop isolator (insertion 0.8 dB, isolation 40 dB); an intracavity erbium amplifier (noise figure 5 dB) compensating the passive chain; a 70/30 output coupler; a 1 GHz photoreceiver (about 98 rungs passed); a probe laser of 100 kHz linewidth; and a piezo fiber stretcher for the loop phase. These values give a loaded ratio $\kappa/\delta = 0.19$, an on-resonance conversion ratio of order unity, and an amplifier-noise-limited single-shot amplitude signal-to-noise ratio of order several hundred. One design tension is worth recording: an amplifier that fully compensates the passive losses raises the finesse and suppresses the frequency-domain visibility of the transcendental lineshape (which scales as $(\kappa/\delta)^2$); heavy output coupling restores κ/δ to the required range while the amplifier still compensates the passive chain. The time-domain protocol below is insensitive to this tension.

4.3 Time-domain protocol

Pulse the clockwise port; record the counter-clockwise impulse response. In the model, this response is a comb of revivals, one per round trip, carried by a secular envelope $t e^{-\kappa t/2}$ — the time-domain fingerprint of the double-pole (Jordan) structure, as opposed to the plain exponential of a diagonalizable pair. Three measurements are proposed. (i) Secular-envelope exponent: fit the revival peak heights to $t^p e^{-\kappa t/2}$; the Jordan structure predicts $p = 1$, a diagonalizable control predicts $p = 0$. (ii) Loop-phase sweep: the revival positions shift linearly with the piezo-controlled loop phase, with slope $1/\delta$; sweeping the phase at fixed ports scans the angular structure of the kernel without moving parts. (iii) Reciprocity test: the one-way configuration predicts a single revival comb; any reciprocal back-coupling produces a twin comb at the mirrored phase. The twin-comb contrast is floored by the finite isolation of the isolator (energy floor $10^{-\text{isolation}/10}$); the test therefore reads the mirror-deletion statement of Section 2.3 directly on an oscilloscope, down to that floor. Figures 1 and 2 illustrate the two signatures on the deterministic model itself. We emphasize the epistemic order: these are proposed measurements; none has been performed.

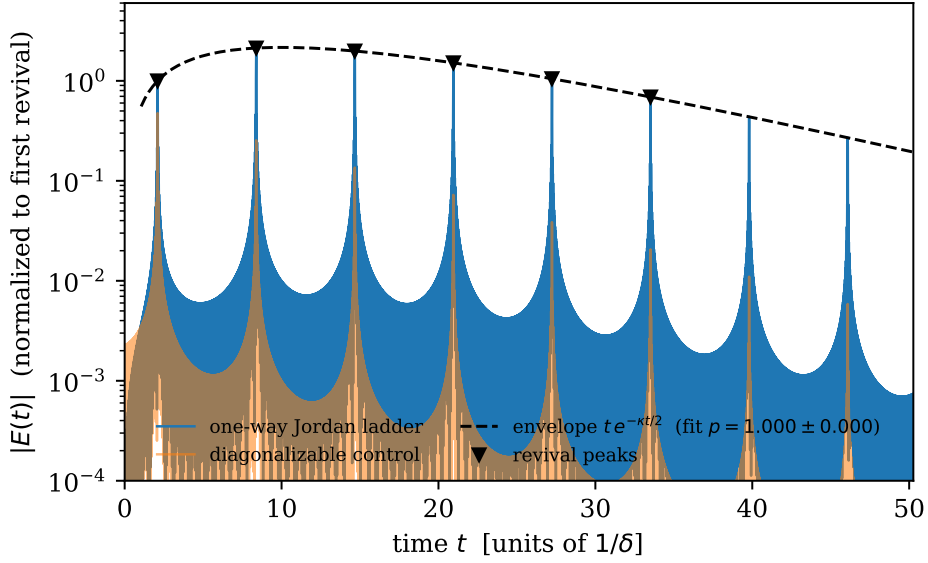


Figure 1: Simulation of the model of Section 2 in the mapped dimensionless regime ($\kappa/\delta = 0.2$, $M = 300$ rungs, loop phase $\phi = 2.1$): revival comb $|E(t)|$ of the one-way Jordan ladder versus the diagonalizable control, with the secular envelope $t e^{-\kappa t/2}$ anchored on the fitted revival peaks. The three-regressor fit of Section 4.3 recovers $p = 1.0001 \pm 0.0001$ (Jordan prediction $p = 1$) and $\hat{\kappa} = 0.2000$. Deterministic model output; no experimental data.

5 Noise and imperfection analysis

5.1 Classification and grouped channels

For this protocol the decisive distinction is between within-record effects (acting inside one 0.8-microsecond record, hence able to distort the secular exponent) and between-shot effects (acting between averaged records, hence damping the revival comb: a shot-to-shot loop-phase spread σ suppresses rung m by $e^{-m^2 \sigma^2/2}$). Grouping ten physical effects by this criterion and by channel yields four families: (A) loop phase noise (thermal, acoustic, vibrational — the fiber thermo-optic response is of order 10^3 radians per kelvin over 20 m, so unstabilized drifts translate into radians over an averaging run); (B) intensity noise (amplifier relative intensity noise within the record; slow gain drift between shots); (C) polarization (state-of-polarization drift converting, through polarization-dependent loss, into slow amplitude fading); (D) parasitic reflections and finite isolation (a genuine weak twin comb at the isolation floor; distributed Rayleigh backscatter sits well below it for a 20 m loop). Kerr nonlinearity and third-order dispersion are included as switches and shown negligible at milliwatt powers (of order 10^{-5} radian per round trip, and 10^{-3} of the second-order dispersion, respectively). All magnitudes are literature-typical and flagged as such.

5.2 Simulation results

With additive detection noise alone, the secular exponent separates from the diagonalizable value at the ten-sigma level in a single shot at amplitude signal-to-noise 10, and at about forty sigma at signal-to-noise 3 with one hundred oscilloscope averages. Replacing white noise by amplifier noise recirculating on the loop resonances — the least favorable spectral shape, since the noise then peaks exactly where the signal lives — degrades this only mildly. The exponent is likewise unaffected by detector-bandwidth truncation down to about fifteen rungs, by slow polarization fading, by gain jitter, and, notably, by shot-to-shot loop-phase jitter: the secular envelope of the revival peaks does not depend

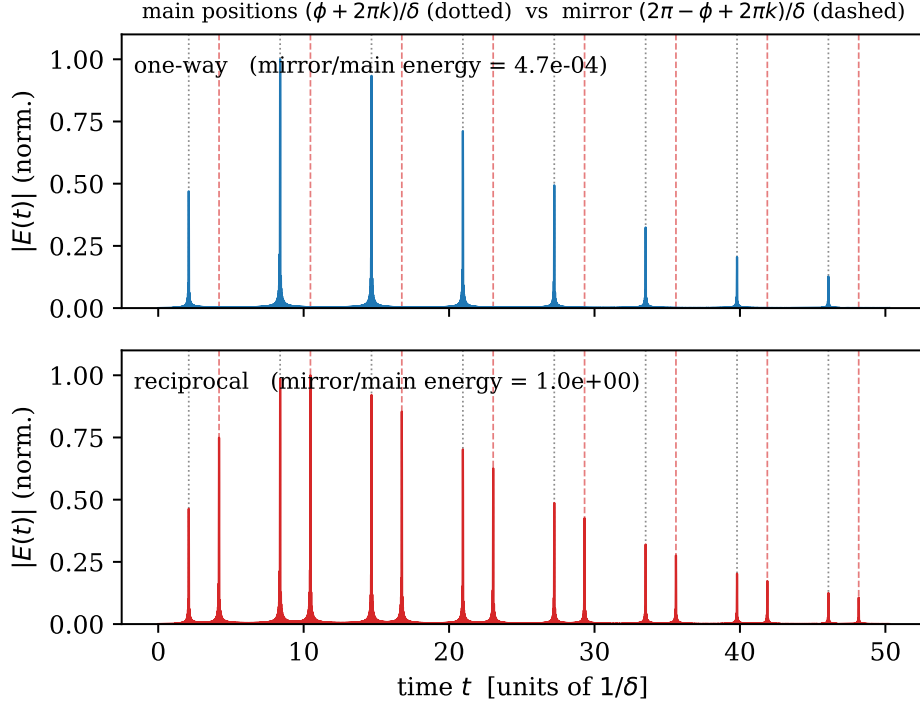


Figure 2: The mirror-deletion law in the time domain (same model and parameters as Fig. 1). Top: one-way configuration — a single revival comb at the main positions $(\phi + 2\pi k)/\delta$ (dotted); the energy at the mirrored positions $(2\pi - \phi + 2\pi k)/\delta$ (dashed) is at the numerical floor (mirror-to-main energy ratio 4.7×10^{-4}). Bottom: reciprocal control — the twin comb appears at the mirrored positions with ratio ≈ 1 . Deterministic model output; no experimental data.

on the ladder size, so the Jordan certificate survives phase noise that visibly destroys the comb. What phase noise does destroy is the many-rung content of the comb — the part of the signal that carries the transcendental class — with an effective ladder m_{eff} of order $1/\sigma$. Loop dispersion is subcritical at the mapped design and produces a clear failure signature (late revivals broadening with revival index) only when exaggerated far beyond it. Finite laser linewidth acts, after ensemble averaging, as an additional decay rate π times the linewidth: it biases the extracted κ but provably not the exponent, whose regressor is separated in the fit. Finally, two facts matter for the reciprocity test: the isolation floor of a standard 40 dB isolator lies below the realistic noise-floor of the twin-comb contrast, so single-stage isolation suffices initially; and loop phase noise pollutes the twin-comb contrast roughly tenfold before it kills the comb, so the reciprocity test demands a phase-quiet loop even more than the comb itself does.

5.3 Scope and ranking

Within the explored window of literature-typical and stressed values, loop phase stability dominates every other imperfection; intensity, polarization and reflection channels are benign. We state the ranking with its exact scope: the damage mechanisms and their scalings in the mode index are established in the model; the ordering for a specific real device depends on the relative magnitudes of its actual imperfections and remains to be established from measured component data. The corresponding design guidance — protect the loop phase first — is conditional on that window.

6 Discussion and limitations

6.1 Modeling assumptions

Four idealizations deserve explicit statement. The mode families are taken as ideal counter-winding ladders with a strictly linear frequency ladder; real devices have dispersion, which we treat perturbatively in Section 5 but which deforms the exact closed form. The one-way coupling is Lindbladian and exact (the reverse element vanishes identically); a physical isolator has finite isolation, which converts the exact statement “the mirror partner is deleted” into the quantitative statement “the mirror partner is suppressed to the isolation floor”. The laser is treated at the ensemble level, where its finite linewidth becomes a deterministic additional decay; individual records carry phase-diffusion realizations that the ensemble treatment averages. Finally, every numerical value in Sections 4–5 is literature- or catalogue-typical; none is a measurement of an actual component.

6.2 What a measurement would and would not establish

Observation of the three signatures — secular exponent, phase-swept revival comb, single-versus-twin comb down to the isolation floor — would establish that the device realizes a two-point transfer kernel of the derived class: exchange-non-Hermitian, with the anti-holomorphic sector structure of Section 2. It would not establish anything beyond that. In particular it would carry no implication for arithmetic questions (Section 3.3), no statement about systems other than the one measured, and no foundational or cosmological consequence. Conversely, the measurement is informative in failure: each signature refutes a specific structural claim, as follows.

6.3 Falsifiability

The proposal is refutable in parts. (i) If the secular envelope shows an exponent consistent with zero under conditions where per-pair degeneracy and one-way coupling are independently verified, the Jordan-ladder structure is refuted. (ii) If the conversion channel of the two-point response depends on the coordinate difference rather than the coordinate sum, the one-way kernel structure is refuted. (iii) If a twin comb appears in the one-way configuration at a contrast far above the independently measured isolation floor, the mirror-deletion statement is refuted. (iv) The converse direction is bounded: none of the three signatures, if observed, establishes more than the kernel structure itself.

6.4 Open questions

Three questions are open at the level of mathematics. First, the exact status of the resemblance to mock-modular completions: whether any map exists under which the exchange operation realizes a modular transformation and the one-way kernel arises as the shadow of a specific mock object; we possess no such map and no argument excluding one. Second, the completeness of the classification: our three formula-level properties separate all cases we have constructed, but we have no theorem that they separate all kernels of physical origin. Third, the boundary behavior: the closed form lives on the open unit polydisk and the proposed measurement reads its boundary values; a rigorous treatment of the boundary limit, in the presence of the complex ladder offset, would strengthen Section 4. A structural characterization of this boundary — its intrinsic, dimensionless location, its emergence under truncation of the mode ladder, and the analytic continuation and branch structure beyond it — is given in Section 2.6; the oriented (chiral) crossing law and the detuning-governed interference relief along the cut are characterized there as well; the rigorous limit statement remains open.

7 Conclusion

We have posed the removability question for anti-holomorphic structure in the setting where, we argue, it is best posed — the two-point transfer kernel of a non-reciprocal open system — and answered it at the level of certified formula structure: one-way coupling between counter-winding mode ladders produces a Lerch-transcendent kernel whose exchange-Hermitian mirror partner is absent, whereas every reciprocal construction we tested retains that partner. We have proposed a fiber-loop measurement of this structure, with an analysis of its realistic imperfections and explicit criteria under which the proposal would be refuted. No experimental data exist; the next step is a measurement.

Code and data availability

The symbolic certification harness, the simulation scripts, and the findings files underlying every certified claim in this paper are archived alongside this deposit. This work extends the author’s eml* program on the detection of anti-holomorphic dynamics; the corresponding paper and software are archived at Zenodo under DOIs [10.5281/zenodo.20272784](https://doi.org/10.5281/zenodo.20272784) (paper) and [10.5281/zenodo.20273262](https://doi.org/10.5281/zenodo.20273262) (software), to which the present deposit is linked.

AI assistance disclosure

The author, an independent researcher, developed this work with AI assistants used as coding, auditing, and drafting tools under a strict verification discipline: every symbolic claim was certified by exact Wirtinger-derivative computation executed on the author’s machine, and no numerical result reported here was produced or accepted without such execution. Responsibility for the content rests with the author.

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